

Alaoglu–Erdős at Depth Two

An Independent Proof Audit, Quadratic-Rank Reduction, and the Exact Remaining Obstruction

Independent continuation report

21 August 2026

Abstract

We investigate the assertion

$$2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}$$

for real x . The supplied unified report already develops a very large collection of semigroup, valuation, determinant, approximation, p -adic, matrix, probability, special function, and formal-verification approaches. To avoid duplicating that work, the present report takes a different route: it isolates the conjecture as a positivity-constrained quadratic relation in the tensor square of the prime-logarithm space and studies the *rank* of that relation.

The main unconditional advance is a low-rank exclusion theorem. Baker’s homogeneous linear-independence theorem implies that a nonzero rational quadratic form of rank at most two cannot vanish at a tuple of $\mathbb{Q}$ -linearly independent logarithms of positive algebraic numbers. Applied to Alaoglu–Erdős, this proves that every hypothetical counterexample must generate a genuinely rank-three or rank-four quadratic period relation. We classify the two possibilities exactly. In multiplicative rank three, the relation is a smooth rational conic with a distinguished rational point, and a rational column operation symmetrizes the singular logarithm matrix. Consequently a counterexample in this branch would produce three multiplicatively independent positive algebraic numbers whose logarithms form a geometric progression; more precisely, for $\theta = \log 3 / \log 2$ there would exist $\alpha > 1$ algebraic such that α , α^θ , and α^{θ^2} are all algebraic. In multiplicative rank four, the associated two-star quadratic has full rank four.

These reductions are rigorous, but they do not constitute an unconditional proof of the conjecture. The desired conclusion is an exact 2×2 case of the Four Exponentials Conjecture, equivalently of the 2×2 Matrix Coefficient Conjecture. The latter was published in 2026 as a conjectural framework, with a key implication still left open. The report therefore states plainly which steps are theorems, which are equivalences, and which single depth-two nonvanishing statement is still missing. It also explains why recent complete classifications of *linear* relations among 1-periods do not decide this quadratic product relation.

Audit conclusion

No unconditional proof of the Alaoglu–Erdős conjecture is presently obtained here, and no such proof is claimed. An assertion to the contrary would conceal an unproved instance of Four Exponentials (or an equivalent matrix-coefficient statement). What is proved is a new, exact reduction: a counterexample can exist only through a rank-three or rank-four quadratic relation among independent algebraic logarithms; all rank-one and rank-two mechanisms are ruled out unconditionally by Baker’s theorem.

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1 Scope, nonduplication, and theorem ledger

1.1 What this report does not repeat

The attached unified report is already encyclopedic. It includes, among many other themes, the logarithmic determinant, the Four Exponentials reduction, exact conditional structure of the solution monoid, prime-support and valuation lattices, rational-function rigidity, finite-difference and interpolation arguments, determinant divisibility, p -adic and tropical routes, matrix-coefficient formulations, point-counting strategies, probabilistic models, special-function approaches, and a substantial Lean corpus.

Repeating those developments would add length without adding mathematical information. The present continuation therefore deliberately omits their proofs and develops four pieces that are logically and technically different:

- (i) a degree-two period map on the formal prime-exponent space;
- (ii) a Baker-theoretic exclusion of every quadratic relation of rank at most two;
- (iii) a hyperbolic pullback classification that identifies the only remaining ranks;
- (iv) a rational symmetrization theorem for the multiplicatively dependent branch.

The resulting report is self-contained for these claims and uses the broad earlier report only as a reason to keep the scope focused.

1.2 Ledger of claims

For ease of audit, the logical status of the principal statements is recorded in [Table 1](#).

Statement	Status	Content
Elementary reduction	Theorem	A nonintegral solution is positive, irrational, and in fact transcendental.
Four Exponentials reduction	Exact implication	Four Exponentials immediately rules out a nonintegral solution.
Restricted matrix coefficient	Exact equivalence at the relevant matrix	A counterexample gives a singular logarithm matrix with no vanishing rational matrix coefficient.
Prime-log two-star formulation	Exact equivalence	The conjecture is injectivity of a degree-two period map on a small positive cone.
Baker rank-two exclusion	Theorem	No nonzero rational quadratic form of rank ≤ 2 vanishes on independent algebraic logarithms.
Counterexample rank dichotomy	Theorem	The formal two-star quadratic of a counterexample has rank exactly 3 or 4.
Rank-three conic form	Theorem	The dependent branch is a nondegenerate split ternary conic with an explicit determinant and parametrization.
Symmetric logarithm normal form	Theorem	The dependent branch rationally symmetrizes to three independent logarithms in geometric progression.

Statement	Status	Content
Depth-two nonvanishing	Open	Exclude the remaining rank-three and rank-four two-star relations.
Alaoglu–Erdős	Open in this report	Follows from Four Exponentials, Matrix Coefficient, Structural Rank, or Schanuel.

Table 1: Logical status of the main assertions.

2 Elementary reductions and the singular logarithm matrix

Throughout, a *candidate* is a triple (x, M, N) with

$$x \in \mathbb{R}, \quad M = 2^x \in \mathbb{N}_{>0}, \quad N = 3^x \in \mathbb{N}_{>0}.$$

A candidate is a *counterexample* if $x \notin \mathbb{Z}$.

2.1 The rational and algebraic cases

Proposition 2.1 (Elementary boundary). *Let (x, M, N) be a candidate.*

- (a) $x \geq 0$.
- (b) If $x \in \mathbb{Q}$, then $x \in \mathbb{N}_0$.
- (c) If x is algebraic, then $x \in \mathbb{N}_0$.
- (d) Hence every counterexample has $x > 0$ transcendental.

Proof. If $x < 0$, then $0 < 2^x < 1$, contradicting $2^x \in \mathbb{Z}$. Thus $x \geq 0$. Write a rational $x = a/b$ in lowest terms with $b > 0$. The identity $2^a = M^b$ and unique factorization give $a = b \operatorname{ord}_2(M)$, so $b \mid a$. Coprimality forces $b = 1$, proving (b). If x is algebraic and irrational, the Gelfond–Schneider theorem makes 2^x transcendental, contrary to $M \in \mathbb{Z}$. Thus algebraic x is rational and (b) applies. The last claim is immediate. $\square$

The same unique-factorization argument shows that $\log 2$ and $\log 3$ are $\mathbb{Q}$ -linearly independent: a relation $a \log 2 + b \log 3 = 0$ with integers a, b would exponentiate to $2^a 3^b = 1$, forcing $a = b = 0$.

2.2 The rank-one real matrix and the rank-two formal problem

Put

$$u = \log 2, \quad v = \log 3,$$

and form

$$L = L(M, N) = \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log N \end{pmatrix}. \tag{1}$$

For a candidate,

$$L = \begin{pmatrix} u \\ v \end{pmatrix} \begin{pmatrix} 1 & x \end{pmatrix}, \quad \det L = u \log N - v \log M = 0. \tag{2}$$

Thus the matrix has real rank one. Its entries are logarithms of positive algebraic numbers. The problem is that their *formal* dependence over $\mathbb{Q}$ need not have rank one.

Proposition 2.2 (Four Exponentials implication). *The Four Exponentials Conjecture implies the Alaoglu–Erdős conjecture.*

Proof. Suppose x were a counterexample. The pair u, v is $\mathbb{Q}$ -linearly independent, and so is $1, x$. Four Exponentials applied to the two rows u, v and the two columns $1, x$ asserts that at least one of

$$e^u = 2, \quad e^{ux} = M, \quad e^v = 3, \quad e^{vx} = N$$

is transcendental. All four are positive integers, a contradiction. $\square$

2.3 The exact matrix-coefficient obstruction

For nonzero rational column vectors $w, z \in \mathbb{Q}^2$,

$$w^T L z = (w_1 u + w_2 v)(z_1 + x z_2). \quad (3)$$

This elementary factorization is the sharpest possible audit tool.

Proposition 2.3 (Rational anisotropy). *If (x, M, N) is a counterexample, then*

$$w^T L z \neq 0 \quad \text{for every } w, z \in \mathbb{Q}^2 \setminus \{0\}.$$

Proof. The first factor in (3) is nonzero because u, v are $\mathbb{Q}$ -linearly independent. The second is nonzero because a relation $z_1 + x z_2 = 0$ with rational z_1, z_2 , not both zero, would make x rational. $\square$

The 2×2 Matrix Coefficient Conjecture says that a singular matrix of logarithms of algebraic numbers has such a vanishing rational matrix coefficient. Therefore [Theorem 2.3](#) says exactly that a counterexample would violate that conjecture. Dasgupta and Kakde prove that the 2×2 Matrix Coefficient Conjecture is equivalent to Four Exponentials and that the stronger Structural Rank Conjecture implies it; their paper leaves open a key implication in the proposed proof strategy [5]. This is not a historical side note: it identifies the precise contemporary frontier occupied by (1).

Exact frontier

A proof that starts only from the facts “ L is singular” and “the entries of L are logarithms of algebraic numbers” must supply, in this special case, the missing conclusion of the Matrix Coefficient Conjecture. The factorization (3) then finishes the argument instantly.

3 The prime-exponent space and a depth-two period map

3.1 Formal prime logarithms

Let $\mathcal{P}$ be the set of primes and let

$$V = \bigoplus_{p \in \mathcal{P}} \mathbb{Q} e_p$$

be the rational vector space with one basis vector for each prime. For a positive rational number r , define its exponent vector

$$\nu(r) = \sum_{p \in \mathcal{P}} \text{ord}_p(r) e_p \in V.$$

Define a linear evaluation map

$$\ell : V \longrightarrow \mathbb{R}, \quad \ell(e_p) = \log p. \quad (4)$$

Then $\ell(\nu(r)) = \log r$.

Lemma 3.1 (Unique-factorization injectivity). *The map ℓ is injective.*

Proof. If $\sum_p c_p \log p = 0$ with finitely many $c_p \in \mathbb{Q}$, clear denominators to obtain integers n_p with $\sum_p n_p \log p = 0$. Exponentiation gives $\prod_p p^{n_p} = 1$, and unique factorization forces every $n_p = 0$. $\square$

The first symmetric power V records linear logarithmic relations. The determinant condition lies in the second symmetric power. Extend (4) multiplicatively to

$$\pi_2 : \text{Sym}^2(V) \longrightarrow \mathbb{R}, \quad \pi_2(rs) = \ell(r)\ell(s) \quad (r, s \in V). \quad (5)$$

This is the degree-two, or “depth-two,” period map used in this report. The word “depth” is local terminology: it simply distinguishes products of logarithms from linear relations among logarithms.

3.2 The two-star tensor

Let

$$a = \nu(M), \quad b = \nu(N).$$

Define

$$q_{a,b} = e_2 b - e_3 a \in \text{Sym}^2(V). \quad (6)$$

It has a graph-theoretic shape consisting of two stars, one centered at 2 and one centered at 3, with opposite signs. Evaluation gives

$$\pi_2(q_{a,b}) = \log 2 \log N - \log 3 \log M. \quad (7)$$

Thus every candidate satisfies $\pi_2(q_{a,b}) = 0$.

To describe the formal zero, write

$$a = \sum_p a_p e_p, \quad b = \sum_p b_p e_p.$$

The coefficient of $e_2 e_p$ for $p \notin \{2, 3\}$ is b_p , the coefficient of $e_3 e_p$ is $-a_p$, the coefficients of e_2^2, e_3^2 are $b_2, -a_3$, and the coefficient of $e_2 e_3$ is $b_3 - a_2$.

Lemma 3.2 (Formal zero classification). *For $a, b \in \bigoplus_p \mathbb{N}_0 e_p$, one has $q_{a,b} = 0$ in $\text{Sym}^2(V)$ if and only if there is $m \in \mathbb{N}_0$ such that*

$$a = m e_2, \quad b = m e_3.$$

Proof. If $q_{a,b} = 0$, the coefficients just listed show that $b_p = 0$ for $p \neq 3$, that $a_p = 0$ for $p \neq 2$, and that $b_3 = a_2$. Calling the common value m gives the stated form. The converse is immediate. $\square$

Define the positive two-star cone

$$\mathcal{C}_{2,3} = \{e_2 b - e_3 a : a, b \in \bigoplus_p \mathbb{N}_0 e_p\} \subset \text{Sym}^2(V). \quad (8)$$

Theorem 3.3 (Exact depth-two formulation). *The following are equivalent.*

- (i) *For every real x , $2^x, 3^x \in \mathbb{Z}$ implies $x \in \mathbb{Z}$.*
- (ii) $\ker \pi_2 \cap \mathcal{C}_{2,3} = \{0\}$.
- (iii) *For all nonnegative integral exponent vectors a, b , the equality*

$$\log 2 \ell(b) = \log 3 \ell(a)$$

forces $a = m e_2, b = m e_3$ for some $m \in \mathbb{N}_0$.

Proof. The equivalence of (ii) and (iii) is [Theorem 3.2](#). Suppose (ii), and let (x, M, N) be a candidate. By (7), $q_{a,b} \in \ker \pi_2$; hence it is formally zero, so $M = 2^m$, $N = 3^m$. Since $2^x = 2^m$, one has $x = m$. Conversely, if (ii) fails, choose nonnegative integral a, b with nonzero $q_{a,b}$ and $\pi_2(q_{a,b}) = 0$. Put $M = \prod p^{a_p}$ and $N = \prod p^{b_p}$. Then

$$x = \frac{\log M}{\log 2} = \frac{\log N}{\log 3}$$

gives $2^x = M$ and $3^x = N$, while [Theorem 3.2](#) shows $x \notin \mathbb{Z}$. □

Exact frontier

The conjecture is not asking for all quadratic relations among prime logarithms to vanish. It asks only for injectivity of π_2 on the extremely sparse, positivity-constrained cone $\mathcal{C}_{2,3}$. This is much weaker than algebraic independence of prime logarithms, but it is genuinely quadratic and therefore lies beyond Baker’s linear theorem.

4 Baker’s theorem eliminates quadratic rank at most two

4.1 Rank of a quadratic tensor

If U is a finite-dimensional rational vector space and $q \in \text{Sym}^2(U)$, contraction makes q a symmetric linear map

$$q^\sharp : U^* \longrightarrow U.$$

Its rank is the *rank* of q . In coordinates this is the rank of any symmetric coefficient matrix for the quadratic form, and is independent of the harmless convention of putting factors 2 on mixed terms.

We use the following qualitative form of Baker’s theorem.

Theorem 4.1 (Baker, homogeneous form). *Let $\lambda_1, \dots, \lambda_n$ be logarithms of nonzero algebraic numbers. If they are linearly independent over $\mathbb{Q}$, then they are linearly independent over $\overline{\mathbb{Q}}$.*

For positive algebraic numbers, the real logarithms remove all branch issues.

Lemma 4.2 (Low-rank factorization). *Let K be an algebraically closed field of characteristic different from 2. Every nonzero quadratic form over K of rank one or two is a product of two nonzero linear forms over K .*

Proof. After an invertible linear change of variables, a rank-one form is X_1^2 . A rank-two form is $X_1^2 + X_2^2$, which factors over the algebraically closed field as $(X_1 + iX_2)(X_1 - iX_2)$ after choosing $i^2 = -1$. □

Theorem 4.3 (Baker rank-two exclusion). *Let $\lambda_1, \dots, \lambda_n$ be $\mathbb{Q}$ -linearly independent logarithms of positive algebraic numbers. If $0 \neq q \in \mathbb{Q}[X_1, \dots, X_n]$ is homogeneous quadratic and $\text{rank}(q) \leq 2$, then*

$$q(\lambda_1, \dots, \lambda_n) \neq 0.$$

Proof. By [Theorem 4.2](#), over $\overline{\mathbb{Q}}$ one may write $q = L_1 L_2$, where L_1, L_2 are nonzero algebraic linear forms; in rank one they may coincide. If $q(\lambda) = 0$, then $L_1(\lambda) = 0$ or $L_2(\lambda) = 0$. This is a nontrivial $\overline{\mathbb{Q}}$ -linear relation among the λ_i , contradicting [Theorem 4.1](#). □

New unconditional result

Theorem 4.3 is a genuine quadratic consequence of Baker’s linear independence theorem. It does not prove that arbitrary quadratic forms are nonzero; it identifies an exact algebraic boundary. Rank one and rank two factor into linear forms and are therefore impossible. Rank three is the first case in which factorization no longer reduces the problem to Baker.

4.2 Application to the two-star tensor

Let $U = \text{span}_{\mathbb{Q}}\{e_2, e_3, a, b\}$. Choose a rational basis $f_1, \dots, f_r$ of U . The numbers $\ell(f_i)$ are logarithms of positive algebraic numbers: after clearing denominators, each f_i is an integral combination of prime basis vectors, and division by the common denominator corresponds to a positive algebraic root. They are $\mathbb{Q}$ -linearly independent by Theorem 3.1.

Corollary 4.4 (First unconditional rank restriction). *If (x, M, N) is a counterexample, then the quadratic tensor $q_{a,b}$ has rank at least three.*

Proof. It is nonzero by Theorem 3.2, and its evaluation at the independent algebraic logarithms $\ell(f_i)$ vanishes by (7). Theorem 4.3 excludes rank at most two. $\square$

This restriction is stronger than the familiar statement that a counterexample must involve some prime outside $\{2, 3\}$. It says that the entire quadratic tensor cannot secretly collapse to a product of two algebraic linear forms.

5 Hyperbolic pullback and the exact rank classification

5.1 A four-vector theorem

The special shape $eb - fa$ admits a complete linear-algebra analysis.

Theorem 5.1 (Hyperbolic pullback classification). *Let V be a rational vector space, let $e, f \in V$ be linearly independent, and let $a, b \in V$. Put*

$$q = eb - fa \in \text{Sym}^2(V), \quad r = \dim_{\mathbb{Q}} \text{span}\{e, b, f, a\}.$$

Then:

- (i) *if $r = 4$, then $\text{rank}(q) = 4$;*
- (ii) *if $r \leq 2$, then $\text{rank}(q) \leq 2$;*
- (iii) *if $r = 3$, let*

$$c_1e + c_2b + c_3f + c_4a = 0 \tag{9}$$

be a nonzero relation, unique up to scale. Then

$$\text{rank}(q) = \begin{cases} 2, & c_1c_2 - c_3c_4 = 0, \\ 3, & c_1c_2 - c_3c_4 \neq 0. \end{cases}$$

Proof. Let $E = \mathbb{Q}^4$ with basis z_1, z_2, z_3, z_4 and let $\phi : E \rightarrow \text{span}\{e, b, f, a\}$ send the basis to e, b, f, a . The tensor q is the image of

$$q_0 = z_1z_2 - z_3z_4.$$

As a bilinear form on E^* , twice q_0 has matrix

$$G = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad G^{-1} = G.$$

The bilinear form defined by q is the restriction of this hyperbolic form to the r -dimensional subspace $\text{im}(\phi^*) \subset E^*$.

If $r = 4$, the map ϕ is an isomorphism and the rank is 4. If $r \leq 2$, restriction to an at most two-dimensional space has rank at most 2.

Now suppose $r = 3$. The kernel of ϕ is the line spanned by $c = (c_1, c_2, c_3, c_4)^T$, and

$$\text{im}(\phi^*) = c^\perp = \{z \in E^* : c^T z = 0\}.$$

The restriction of G to $c^\perp$ is degenerate precisely when its G -normal $G^{-1}c = Gc$ also lies in $c^\perp$, equivalently when

$$c^T G^{-1}c = 2(c_1c_2 - c_3c_4) = 0.$$

In that case the radical is one-dimensional and the restricted rank is 2; otherwise the restriction is nondegenerate of rank 3. $\square$

The scalar

$$\Delta_{\text{tan}}(c) = c_1c_2 - c_3c_4 \tag{10}$$

can be viewed as a tangency invariant: the relation hyperplane is tangent to the hyperbolic quadric $z_1z_2 - z_3z_4 = 0$ exactly when $\Delta_{\text{tan}}(c) = 0$.

5.2 Multiplicative rank of the four integers

Let

$$r_\times(2, 3, M, N) = \dim_{\mathbb{Q}} \text{span}\{e_2, e_3, a, b\}. \tag{11}$$

This is the rational multiplicative rank of the subgroup generated by the four positive integers. It is either 2, 3, or 4 because e_2, e_3 are independent.

Theorem 5.2 (Counterexample rank theorem). *For a counterexample (x, M, N) ,*

$$r_\times(2, 3, M, N) \in \{3, 4\}, \quad \text{rank}(q_{a,b}) = r_\times(2, 3, M, N).$$

More explicitly:

- (a) *in multiplicative rank four, $q_{a,b}$ has rank four;*
- (b) *in multiplicative rank three, $q_{a,b}$ has rank three;*
- (c) *multiplicative rank two is impossible.*

Proof. If the multiplicative rank is four, [Theorem 5.1](#) gives rank four. If it is at most two, the same theorem gives quadratic rank at most two, contradicting [Theorem 4.4](#). In multiplicative rank three the quadratic rank is two or three by [Theorem 5.1](#), and Baker excludes rank two. $\square$

New unconditional result

A hypothetical counterexample cannot be explained by any hidden low-dimensional factorization. Its formal quadratic relation has exactly the same rank as the multiplicative group generated by $2, 3, M, N$, and that common rank is 3 or 4.

5.3 The unique relation in rank three

Suppose the multiplicative rank is three. Then there is a unique rational relation, up to scale,

$$M^\alpha N^\beta 2^\gamma 3^\delta = 1, \quad (\alpha, \beta, \gamma, \delta) \in \mathbb{Z}^4 \setminus \{0\}. \quad (12)$$

In exponent space,

$$\gamma e_2 + \beta b + \delta e_3 + \alpha a = 0.$$

Thus the vector c in (9), for the ordering $(e, b, f, a) = (e_2, b, e_3, a)$, is

$$c = (\gamma, \beta, \delta, \alpha).$$

The tangent invariant is

$$\Delta_{\text{tan}} = \gamma\beta - \delta\alpha. \quad (13)$$

Corollary 5.3 (Non-tangency). *For a rank-three counterexample,*

$$\gamma\beta - \delta\alpha \neq 0.$$

Proof. If it vanished, Theorem 5.1 would give $\text{rank}(q_{a,b}) = 2$, contradicting Theorem 5.2. $\square$

There is also a direct interpretation. Taking logarithms in (12) and substituting $\log M = x \log 2$, $\log N = x \log 3$ gives

$$x(\alpha + \beta\theta) + (\gamma + \delta\theta) = 0, \quad \theta = \frac{\log 3}{\log 2}.$$

If $\gamma\beta - \delta\alpha = 0$, numerator and denominator are rationally proportional, forcing $x \in \mathbb{Q}$, which has already been excluded. The rank calculation shows that this familiar fractional-linear determinant is exactly the dual hyperbolic tangency invariant.

6 The rank-three conic and its explicit arithmetic normal form

6.1 A common external prime direction

Decompose

$$V = \text{span}\{e_2, e_3\} \oplus V_{\text{ext}}, \quad V_{\text{ext}} = \bigoplus_{p \neq 2,3} \mathbb{Q}e_p.$$

Write $a_{\text{ext}}, b_{\text{ext}}$ for the projections of a, b . Then

$$r_{\times}(2, 3, M, N) = 2 + \dim_{\mathbb{Q}} \text{span}\{a_{\text{ext}}, b_{\text{ext}}\}. \quad (14)$$

Consequently the rank-three branch is exactly the case in which the two external exponent vectors span one rational line.

After choosing a primitive nonnegative integral vector $h \in V_{\text{ext}}$, there are $u, v \in \mathbb{N}_0$, not both zero, and nonnegative integers A, B, C, D such that

$$\begin{aligned} M &= 2^A 3^B R^u, \\ N &= 2^C 3^D R^v, \end{aligned} \quad R = \prod_{p \neq 2,3} p^{h_p} > 1. \quad (15)$$

One of u, v may vanish, corresponding to one output having no external prime factor.

Put

$$X = \log 2, \quad Y = \log 3, \quad Z = \log R.$$

The determinant relation becomes

$$Q(X, Y, Z) = CX^2 + (D - A)XY - BY^2 + vXZ - uYZ = 0. \quad (16)$$

The three logarithms X, Y, Z are $\mathbb{Q}$ -linearly independent by unique factorization.

6.2 The determinant certificate

Twice the symmetric matrix of (16) is

$$H = \begin{pmatrix} 2C & D - A & v \\ D - A & -2B & -u \\ v & -u & 0 \end{pmatrix}. \quad (17)$$

A direct calculation gives

$$\det H = 2((A - D)uv + Bv^2 - Cu^2). \quad (18)$$

Proposition 6.1 (Rank-three boundary certificate). *A rank-three counterexample represented as in (15) must satisfy*

$$(A - D)uv + Bv^2 - Cu^2 \neq 0. \quad (19)$$

Equivalently, the ternary quadratic form (16) is nondegenerate.

Proof. The rank of Q is the rank of H . Theorem 5.2 says that this rank is three, so the determinant cannot vanish. $\square$

The form Q nevertheless has the rational isotropic point $[0 : 0 : 1]$. Its conic is therefore split and rationally parametrizable. The point is smooth because the gradient there is $(v, -u, 0)$, which is nonzero.

6.3 Parametrization at the prime-log point

Set

$$\theta = \frac{Y}{X} = \log_2 3, \quad \rho = \frac{Z}{X} = \log_2 R.$$

Dividing (16) by X^2 gives

$$C + (D - A)\theta - B\theta^2 + (v - u\theta)\rho = 0. \quad (20)$$

Since θ is irrational and u, v are rational, not both zero, $v - u\theta \neq 0$. Hence

$$\rho = \frac{B\theta^2 - (D - A)\theta - C}{v - u\theta}. \quad (21)$$

Proposition 6.2 (Explicit rank-three obstruction). *Every rank-three counterexample produces integers A, B, C, D, u, v satisfying (19) and an integer $R > 1$, supported away from 2 and 3, for which (21) holds with $\theta = \log_2 3$ and $\rho = \log_2 R$. Conversely, any such data with (20) gives a vanishing two-star quadratic of multiplicative rank three; it gives an Alaoglu–Erdős counterexample exactly when the corresponding ratios*

$$\frac{\log(2^A 3^B R^u)}{\log 2} \quad \text{and} \quad \frac{\log(2^C 3^D R^v)}{\log 3}$$

are equal and nonintegral, which is precisely (20) plus nonintegrality.

The formula is a useful compression of the dependent branch. It does not prove nonexistence: proving that $2^{F(\log_2 3)}$ is transcendental for the relevant nonlinear rational functions F is itself beyond current theorems and contains a Four Exponentials type difficulty.

7 Rational symmetrization and a geometric progression of logarithms

The conic description is coordinate-dependent. A more invariant normal form comes directly from the 2×2 logarithm matrix.

7.1 Symmetrizing a rank-three singular logarithm matrix

For $C, L \in M_2(\mathbb{R})$ use the Frobenius pairing

$$\langle C, L \rangle_F = \text{tr}(C^T L) = \sum_{i,j} C_{ij} L_{ij}.$$

Let

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Theorem 7.1 (Rational symmetrization). *Let $L \in M_2(\mathbb{R})$ be nonzero and singular. Assume that its four entries span a three-dimensional $\mathbb{Q}$ -vector space and that the unique rational relation among them has an invertible coefficient matrix $C \in M_2(\mathbb{Q})$:*

$$\langle C, L \rangle_F = 0, \quad \det C \neq 0.$$

Then there is $Q \in \text{GL}_2(\mathbb{Q})$ such that $S = LQ$ is symmetric. If, in addition, $L = r(1, x)$ is an outer product with $r = (r_1, r_2)^T$, then

$$S = \kappa \begin{pmatrix} r_1^2 & r_1 r_2 \\ r_1 r_2 & r_2^2 \end{pmatrix} \tag{22}$$

for some nonzero real κ .

Proof. Under the right change $S = LQ$, the relation matrix becomes $C' = CQ^{-T}$ because

$$\langle C, L \rangle_F = \langle CQ^{-T}, LQ \rangle_F.$$

Choose Q by $Q^{-T} = C^{-1}J$. Then $C' = J$, and

$$0 = \langle J, S \rangle_F = S_{12} - S_{21},$$

so S is symmetric.

For the second assertion, write $(1, x)Q = d^T$. Then $S = rd^T$. Symmetry says $r_1 d_2 = r_2 d_1$. Since $r \neq 0$ and $d \neq 0$, the vectors d and r are proportional: $d = \kappa r$. This gives (22). $\square$

7.2 Application to a dependent counterexample

For the logarithm matrix (1), a multiplicative relation (12) is

$$\left\langle \begin{pmatrix} \gamma & \alpha \\ \delta & \beta \end{pmatrix}, L \right\rangle_F = 0.$$

Its determinant is $\gamma\beta - \delta\alpha$, which is nonzero by Theorem 5.3. Therefore Theorem 7.1 applies with $r = (u, v)^T = (\log 2, \log 3)^T$.

Theorem 7.2 (Geometric-progression normal form). *Suppose (x, M, N) is a multiplicative-rank-three counterexample. Then there exists $\kappa > 0$ such that the three numbers*

$$\lambda_0 = \kappa(\log 2)^2, \quad \lambda_1 = \kappa \log 2 \log 3, \quad \lambda_2 = \kappa(\log 3)^2 \quad (23)$$

are logarithms of positive algebraic numbers, are $\mathbb{Q}$ -linearly independent, and satisfy

$$\lambda_1^2 = \lambda_0 \lambda_2, \quad \frac{\lambda_1}{\lambda_0} = \frac{\lambda_2}{\lambda_1} = \theta = \frac{\log 3}{\log 2}. \quad (24)$$

Equivalently, there exists a positive algebraic number $\alpha > 1$ such that

$$\alpha, \quad \alpha^\theta, \quad \alpha^{\theta^2} \quad \text{are all algebraic and multiplicatively independent.} \quad (25)$$

Proof. The preceding discussion gives $Q \in \text{GL}_2(\mathbb{Q})$ and nonzero κ with

$$LQ = \kappa \begin{pmatrix} u^2 & uv \\ uv & v^2 \end{pmatrix}.$$

Replacing Q by $-Q$ if necessary makes $\kappa > 0$. Each entry of LQ is a rational linear combination of logarithms of positive integers. Its exponential is therefore a positive algebraic number: rational coefficients correspond to products, quotients, and positive real algebraic roots. This proves algebraicity of the exponentials of the three numbers in (23).

Right multiplication by an invertible rational matrix preserves the $\mathbb{Q}$ -span of all matrix entries. In multiplicative rank three, the entries of L span dimension three. The symmetric matrix LQ has only the three displayed distinct entries, so they are $\mathbb{Q}$ -linearly independent. The geometric-progression identities are immediate.

Set $\alpha = e^{\lambda_0} > 1$. Then $e^{\lambda_1} = \alpha^\theta$ and $e^{\lambda_2} = \alpha^{\theta^2}$ are algebraic. Independence of the logarithms is exactly multiplicative independence of these three algebraic numbers. $\square$

Corollary 7.3 (The common ratio is necessarily transcendental). *In the situation of Theorem 7.2, the common ratio θ cannot be algebraic.*

Proof. It is not rational because $\log 2$ and $\log 3$ are $\mathbb{Q}$ -linearly independent. If it were algebraic irrational, Gelfond–Schneider applied to the algebraic base $\alpha \neq 0, 1$ would make α^θ transcendental, contradicting (25). $\square$

Of course the transcendence of $\theta = \log_2 3$ is already classical by the same theorem, because $2^\theta = 3$. The point of Theorem 7.2 is not to reprove that fact; it is to show that the entire multiplicatively dependent branch of Alaoglu–Erdős has a single symmetric shape.

New unconditional result

The rank-three branch is forced into a three-term algebraic orbit $\alpha, \alpha^\theta, \alpha^{\theta^2}$. This is a smaller and more symmetric obstruction than the original four entries $2, 3, M, N$, but it is still a Four Exponentials type obstruction: current transcendence theorems do not exclude a transcendental exponent θ from having two consecutive algebraic power iterates.

8 Why the complete theory of linear periods does not settle the problem

8.1 Degree one versus degree two

Each $\log \alpha$ for algebraic α is a classical 1-period. Baker’s theorem is a complete rigidity statement for a fundamental class of *linear* relations. The modern work of Huber and Wüstholz gives a conceptual classification of linear relations among 1-periods via 1-motives [7]; recent work makes this picture more complete and, in important formulations, effective [8, 13].

The Alaoglu–Erdős determinant is not a linear relation among the four logarithms. It is

$$(\log 2)(\log N) - (\log 3)(\log M) = 0,$$

a linear relation between *products* of logarithms. In the notation of Section 3, it lies in $\text{Sym}^2(V)$ and asks about $\ker \pi_2$, not $\ker \ell$. Products of periods are again periods, but a complete description of their quadratic relations would be a much deeper period statement.

Proposition 8.1 (Linear-period completeness cannot be applied verbatim). *Any theorem whose conclusion controls only $\overline{\mathbb{Q}}$ -linear relations among the individual numbers*

$$\log 2, \log 3, \log M, \log N$$

does not by itself exclude the determinant relation

$$\log 2 \log N - \log 3 \log M = 0.$$

Proof. A counterexample may have the four logarithms $\overline{\mathbb{Q}}$ -linearly independent in multiplicative rank four, by Baker’s theorem, while still satisfying the displayed quadratic relation. Linear independence in a vector space does not imply linear independence of all products in its symmetric square. In multiplicative rank three there is one rational linear relation, but after choosing a basis of three independent logs the determinant becomes a nondegenerate rank-three quadratic form, again outside the scope of a theorem about linear relations. $\square$

8.2 A torus-homomorphism no-go lemma

The analytic subgroup theorem is extraordinarily powerful for linear logarithmic relations. It is important, however, not to assign to it a nonlinear conclusion that its functorial input does not encode.

Lemma 8.2 (Differentials of torus homomorphisms are integral-linear). *Every algebraic-group homomorphism $\varphi : \mathbb{G}_m^r \rightarrow \mathbb{G}_m^s$ has the form*

$$\varphi(z_1, \dots, z_r)_j = \prod_{i=1}^r z_i^{n_{ji}} \quad (n_{ji} \in \mathbb{Z}),$$

and its differential at the identity sends logarithmic coordinates $\lambda \in \mathbb{C}^r$ to $N\lambda$, where $N = (n_{ji})$.

Proof. The character group of $\mathbb{G}_m^r$ is $\mathbb{Z}^r$. Pullback of each coordinate character of $\mathbb{G}_m^s$ is therefore an integral monomial character on $\mathbb{G}_m^r$, giving the displayed form. Differentiating the monomials at the identity gives the integer matrix N . $\square$

Corollary 8.3 (No automatic linearization of the determinant). *Constructions using only products of algebraic tori, algebraic subgroups, quotients, and algebraic homomorphisms transform logarithmic coordinates by rational or integral linear maps. They do not turn the quadratic tensor $e_2b - e_3a$ into a linear logarithmic relation unless an additional genuinely degree-two identity is supplied.*

This is a scope statement, not a claim that motivic methods can never solve the problem. Tensor products of motives and stronger period conjectures are precisely mechanisms by which degree-two information might enter. The point is that the already-complete degree-one classification is not the missing theorem.

8.3 Conditional period consequences

The following standard conjectural principles all imply the desired nonvanishing:

- Schanuel’s conjecture implies sufficient algebraic independence of independent logarithms;
- the Structural Rank Conjecture implies the Matrix Coefficient Conjecture;
- the Matrix Coefficient Conjecture in size two is equivalent to Four Exponentials;
- injectivity of π_2 on all of $\text{Sym}^2(V)$ is far stronger than needed but immediately settles the positive two-star cone.

The report does not assume any of these conjectures in an unconditional theorem.

9 The 2026 matrix frontier in exact form

9.1 Three matrix statements

Let $\mathcal{L}$ be the $\mathbb{Q}$ -vector space of complex logarithms of algebraic numbers. For a matrix $A \in M_{m,n}(\mathcal{L})$, write

$$A = \sum_{k=1}^r A_k \lambda_k,$$

where the A_k have rational entries and the λ_k are $\mathbb{Q}$ -linearly independent. The relevant conjectures are:

Structural Rank.

The numerical rank of A equals the rank of the generic matrix $\sum A_k X_k$ over $\mathbb{Q}(X_1, \dots, X_r)$.

Matrix Coefficient.

If a square A is singular, there are nonzero rational vectors w, z with $w^T A z = 0$.

Four Exponentials.

If two row parameters and two column parameters are each $\mathbb{Q}$ -linearly independent, at least one of the four corresponding exponentials is transcendental.

Dasgupta and Kakde establish

$$\text{SRC} \implies \text{MCC}_2, \quad \text{MCC}_2 \iff \text{FE}_4 \quad \text{in size 2,}$$

and develop an auxiliary-polynomial strategy, while explicitly leaving a needed implication open [5]. The paper appeared in *Advances in Mathematics* in March 2026.

9.2 Specialization to Alaoglu–Erdős

For a nonintegral candidate, L in (1) is singular. If MCC_2 holds, there are nonzero $w, z \in \mathbb{Q}^2$ with $w^T L z = 0$. Proposition 2.3 says this is impossible. Thus:

Theorem 9.1 (Conditional one-line proof). *Each of the following implies the Alaoglu–Erdős conjecture:*

$$\text{Schanuel} \implies \text{SRC} \implies \text{MCC}_2 \iff \text{FE}_4 \implies \text{AE}_{2,3}.$$

For the final implication, the complete proof is the factorization

$$w^T L z = (w_1 \log 2 + w_2 \log 3)(z_1 + x z_2).$$

The difficult part is not the final line; it is obtaining the vanishing matrix coefficient from singularity.

9.3 Why Six Exponentials stops one row or column short

The Six Exponentials Theorem handles a 3×2 or 2×3 grid. A counterexample supplies only the 2×2 algebraic grid

$$\begin{array}{c|cc} & 1 & x \\ \hline \log 2 & 2 & M \\ \log 3 & 3 & N \end{array}.$$

To add a third row $\log R$, one would need algebraicity of R^x ; to add a third column y , one would need algebraicity of both 2^y and 3^y . Neither is available from the hypotheses. In the rank-three normal form, Six Exponentials similarly sees $\alpha, \alpha^\theta, \alpha^{\theta^2}$ but requires one additional algebraic entry to force a contradiction.

10 Audit of tempting but invalid proof routes

This section records failure modes because the same hidden assumptions repeatedly reappear in attempts at the problem.

10.1 Taking prime valuations of a real power

From $M = 2^x$ one cannot write $\text{ord}_p(M) = x \text{ord}_p(2)$ unless x is already known to be an integer in an algebraic setting where such a valuation identity is defined. The real exponential 2^x is defined analytically; valuation is applied only after the output happens to be an integer. Extending $\text{ord}_p(a^x) = x \text{ord}_p(a)$ to arbitrary real x assumes precisely the arithmetic structure at issue.

10.2 Applying Baker to a product relation

Baker excludes nontrivial algebraic *linear* combinations $\sum \beta_i \log \alpha_i$. The determinant

$$\log 2 \log N - \log 3 \log M$$

has logarithms as coefficients of other logarithms, and those coefficients are generally transcendental. It is not a Baker linear form. The valid consequence of Baker is exactly [Theorem 4.3](#): factorable low-rank quadratics are excluded, while irreducible rank-three and rank-four quadratics remain.

10.3 Treating the exponent as algebraic

Gelfond–Schneider proves the conjecture when x is algebraic. A counterexample, however, would necessarily have transcendental x . Any proof that invokes Gelfond–Schneider on 2^x without first proving algebraicity of x is circular.

10.4 Manufacturing a third base

The Six Exponentials Theorem would solve the problem if there were a third multiplicatively independent algebraic base R with R^x algebraic. The hypotheses give this only for R in the group generated by 2 and 3, which adds no independent row. Choosing an external prime and asserting algebraicity of its x -th power is an extra assumption.

10.5 A noncanonical p -adic exponent

The real identity $2^x = M$ does not canonically define a p -adic number x or a compatible p -adic power 2^x . One may form p -adic logarithms of suitable algebraic numbers after choosing domains and branches, but transferring the same real transcendental exponent into $\mathbb{Q}_p$ requires new data. A proof that silently identifies the real and p -adic exponents has changed the problem.

10.6 Integer gaps and rational approximation

Let p/q be a convergent to a hypothetical irrational x . Then

$$|qx - p| < \frac{1}{q}.$$

The integer gap $|M^q - 2^p| \geq 1$ gives only an exponentially small lower bound on $|qx - p|$. A Baker–Matveev lower bound for

$$q \log M - p \log 2 = (qx - p) \log 2$$

is polynomial in q (with a large exponent depending on the fixed algebraic numbers). This is compatible with the $O(1/q)$ upper bound from continued fractions. Synchronizing the analogous base-three expression recreates the original quadratic determinant rather than producing a contradiction.

10.7 Confusing formal and numerical nonvanishing

The tensor $q_{a,b}$ is formally nonzero for a nonintegral candidate. It does not follow that its specialization $e_p \mapsto \log p$ is nonzero. Proving that implication on the positive two-star cone is exactly [Theorem 3.3](#); on all tensors it would be a much stronger algebraic-independence statement.

10.8 Symmetrization is a reduction, not a proof

The equality

$$(\log \beta)^2 = (\log \alpha)(\log \gamma)$$

for algebraic α, β, γ is not ruled out by Baker when the three logarithms are independent. The symmetric normal form exposes a clean obstruction but does not make it linear.

11 Minimal missing statements

11.1 The exact two-star separation principle

Conjecture 11.1 (Positive two-star separation). *For nonnegative integral exponent vectors a, b ,*

$$\pi_2(e_2b - e_3a) = 0 \implies e_2b - e_3a = 0.$$

By [Theorem 3.3](#), this is not merely sufficient; it is equivalent to the Alaoglu–Erdős conjecture. Its advantage is diagnostic: the positivity and two-star support are explicit, and the only allowed formal zero is visible coefficient by coefficient.

11.2 Rank-separated versions

Let $\mathcal{C}_{2,3}^{(r)}$ be the subset of the two-star cone whose underlying exponent span has dimension r .

Conjecture 11.2 (Rank-three two-star separation). $\ker \pi_2 \cap \mathcal{C}_{2,3}^{(3)} = \{0\}$.

Conjecture 11.3 (Rank-four two-star separation). $\ker \pi_2 \cap \mathcal{C}_{2,3}^{(4)} = \{0\}$.

Theorem 11.4 (Exact branch decomposition). *The Alaoglu–Erdős conjecture is equivalent to the conjunction of [Theorems 11.2](#) and [11.3](#). The rank-three conjecture is equivalent to excluding the nondegenerate ternary conics (16) at the prime-log point; the rank-four conjecture excludes full-rank hyperbolic pullbacks.*

Proof. [Theorem 5.2](#) proves that every counterexample lies in exactly one of the two branches. Conversely, any nonzero element in either kernel intersection constructs a counterexample exactly as in the proof of [Theorem 3.3](#). The conic description is [Section 6](#), and the rank-four description follows from [Theorem 5.1](#). $\square$

11.3 A Baker-completion formulation

Conjecture 11.5 (Two-star rank drop). *If $q = e_2b - e_3a \in \mathcal{C}_{2,3}$ satisfies $\pi_2(q) = 0$, then $\text{rank}(q) \leq 2$.*

Proposition 11.6. *Conjecture 11.5 implies the Alaoglu–Erdős conjecture.*

Proof. If q is a nonzero vanishing two-star tensor, the conjectured rank drop and [Theorem 4.3](#) contradict one another. Thus every vanishing two-star tensor is zero, and [Theorem 3.3](#) applies. $\square$

This formulation isolates the only way Baker could finish the proof: one must force the quadratic relation into the factorable range. [Theorem 5.2](#) shows that a counterexample would do exactly the opposite, retaining rank three or four.

11.4 A symmetric orbit formulation for rank three

Conjecture 11.7 (No algebraic θ -orbit of length three). *Let $\theta = \log_2 3$. There is no algebraic $\alpha > 1$ for which $\alpha, \alpha^\theta, \alpha^{\theta^2}$ are all algebraic and multiplicatively independent.*

Corollary 11.8. *Conjecture 11.7 rules out the multiplicative-rank-three branch of the Alaoglu–Erdős conjecture.*

Proof. This is the contrapositive of [Theorem 7.2](#). □

This is not asserted to settle rank four. It is a sharply targeted three-number problem that captures the entire dependent branch.

12 Research directions suggested by the reduction

12.1 Attack rank three first

The rank-three branch has several extra structures absent in rank four:

- one common external prime direction R ;
- a split nondegenerate ternary conic with a rational point;
- the explicit determinant (18);
- the rational-function relation (21);
- the symmetric orbit $\alpha, \alpha^\theta, \alpha^{\theta^2}$.

A promising theorem would exploit these simultaneously rather than treating an arbitrary Four Exponentials matrix. In particular, any auxiliary-polynomial construction can be specialized to a conic with known rational parametrization and to nonnegative integral support data.

12.2 A tensor-square version of the one-period theory

The modern one-period results explain all linear relations among the first-level periods. The present reduction asks for a highly restricted statement in the tensor square of the Tate logarithm part. One possible program is:

- (1) identify the motivic tensor represented by $e_2b - e_3a$;
- (2) classify functorial or shuffle-type relations that can meet the two-star cone;
- (3) prove that every such motivic relation is already the formal zero of [Theorem 3.2](#);
- (4) use an appropriate period injectivity theorem for this restricted tensor class.

The missing step is not expected to follow from the existing linear classification alone.

12.3 Support-adapted matrix-coefficient constructions

The general Matrix Coefficient Conjecture permits arbitrary logarithm matrices. Here every entry comes from two fixed prime rows and two nonnegative integer exponent columns. In the prime basis, the determinant has only two star centers. An auxiliary polynomial or fewnomial argument that preserves this support may need substantially fewer conditions than the general matrix problem. The target should be a vanishing rational matrix coefficient, not merely another small determinant estimate.

12.4 Formal verification targets

The new linear algebra is well suited to formalization. A compact verified development could include:

- the map π_2 and formal zero classification;
- rank-one/rank-two factorization over an algebraic closure;
- the hyperbolic restriction calculation and tangency invariant;
- the determinant formula (18);
- rational symmetrization by $Q^{-T} = C^{-1}J$.

Formalization would certify the reductions, not the missing transcendence statement.

13 Conclusions

The Alaoglu–Erdős statement is elementary to formulate and extraordinarily rigid, but its exact difficulty is modern transcendence theory rather than elementary divisibility. The investigation in this report yields the following clean picture.

First, any counterexample is a positive transcendental exponent and creates a singular 2×2 matrix of algebraic logarithms with no vanishing rational matrix coefficient. That is exactly the size-two Matrix Coefficient/Four Exponentials obstruction.

Second, in the formal prime-exponent space the determinant is the sparse quadratic tensor $q_{a,b} = e_2b - e_3a$. The conjecture is exactly injectivity of the degree-two period map on the positive two-star cone.

Third, Baker’s theorem goes farther than a purely linear audit might suggest: it excludes every vanishing two-star tensor of rank one or two. Therefore a counterexample must carry a genuine rank-three or rank-four quadratic relation. The hyperbolic pullback theorem classifies the ranks exactly.

Fourth, the rank-three branch admits a new symmetric normal form. It would force an algebraic three-term orbit $\alpha, \alpha^{\log_2 3}, \alpha^{(\log_2 3)^2}$ with multiplicatively independent terms. This isolates a compact special obstruction, but current theorems do not exclude it.

The remaining mathematical task is therefore precise: prove nonvanishing for the rank-three and rank-four positive two-star tensors, or prove the corresponding vanishing matrix coefficient. Doing so would prove the conjecture. This report establishes the reductions and the low-rank theorem, but it does not pretend that the final depth-two period statement has already been proved.

A Coordinate proof of the hyperbolic restriction criterion

For reference, we expand the rank-three calculation in coordinates. Let

$$G = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad c = (c_1, c_2, c_3, c_4)^T.$$

The hyperplane $H = c^\perp$ has dimension three. A vector $z \in H$ belongs to the radical of $G|_H$ exactly when $z^T G h = 0$ for every $h \in H$. With the ordinary coordinate pairing, this means Gz belongs

to the annihilator of H , namely $\mathbb{Q}c$. Since $G^{-1} = G$, such a vector must be $z = tGc$. It lies in H exactly when

$$0 = c^T z = t c^T Gc = 2t(c_1c_2 - c_3c_4).$$

If $c_1c_2 - c_3c_4 \neq 0$, the only radical vector is zero and the restriction has rank three. If it vanishes, Gc is a nonzero radical vector. The radical cannot have dimension more than one because G is nondegenerate and H has codimension one, so the restricted rank is two.

B Derivation of the rank-three determinant

Starting with

$$Q = CX^2 + (D - A)XY - BY^2 + vXZ - uYZ,$$

twice its symmetric matrix is

$$H = \begin{pmatrix} 2C & D - A & v \\ D - A & -2B & -u \\ v & -u & 0 \end{pmatrix}.$$

Expanding along the last row gives

$$\begin{aligned} \det H &= v \det \begin{pmatrix} D - A & v \\ -2B & -u \end{pmatrix} + u \det \begin{pmatrix} 2C & v \\ D - A & -u \end{pmatrix} \\ &= v(-(D - A)u + 2Bv) + u(-2Cu - v(D - A)) \\ &= 2((A - D)uv + Bv^2 - Cu^2). \end{aligned}$$

The point $[0 : 0 : 1]$ lies on $Q = 0$. On the affine chart $X \neq 0$, setting $\theta = Y/X$ and $\rho = Z/X$ gives

$$C + (D - A)\theta - B\theta^2 + (v - u\theta)\rho = 0,$$

which is (21).

C One-page conditional proof under Matrix Coefficient

Assume the 2×2 Matrix Coefficient Conjecture. Let $x \in \mathbb{R}$ satisfy $M = 2^x, N = 3^x \in \mathbb{Z}$.

If $x < 0$, then $0 < M < 1$, impossible. If $x \in \mathbb{Q}$, unique factorization applied to $2^x = M$ gives $x \in \mathbb{Z}$. It remains to exclude irrational x .

Form

$$L = \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log N \end{pmatrix} = \begin{pmatrix} \log 2 \\ \log 3 \end{pmatrix} (1, x).$$

Then $\det L = 0$. By Matrix Coefficient, there are nonzero $w, z \in \mathbb{Q}^2$ such that $w^T Lz = 0$. But

$$w^T Lz = (w_1 \log 2 + w_2 \log 3)(z_1 + xz_2).$$

The first factor is nonzero by unique factorization of 2 and 3. The second is nonzero because x is irrational. This contradiction proves $x \in \mathbb{Z}$.

Every line after the invocation of Matrix Coefficient is elementary. The invocation is the unproved step, equivalent in size two to Four Exponentials.

D Nonduplication map

The following table records the intended separation between the supplied unified report and the present continuation.

Theme already extensively represented in the supplied report	Treatment here
Exact solution monoid, primitive generator, counting, density	Not repeated.
Valuation lattices, radical and smooth-output formulations	Used only for elementary prime-exponent notation; no repetition of the lattice theory.
Determinant divisibility and interpolation constructions	Not repeated.
p -adic, tropical, Mahler, probability, matrix, and special-function routes	Not repeated.
Four Exponentials and restricted Matrix Coefficient reduction	Recalled only as the audited frontier needed to place the new results.
Prime-log product independence as a sufficient condition	Replaced by the finer rank-sensitive Baker theorem, which proves nonvanishing for all rank- ≤ 2 quadratics without assuming general product independence.
Multiplicative dependence/PGL branching	Reinterpreted through the hyperbolic tangency invariant and then rationally symmetrized.
Tower formulations	Not redeveloped; only the new symmetric consequence $\alpha, \alpha^\theta, \alpha^{\theta^2}$ arising from rank-three symmetrization is stated.
Formal Lean corpus	Audited conceptually; no claims of a formal proof of the conjecture.

Table 2: How the present report avoids duplicating the supplied report.

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